

House Price Prediction

Internship Week 1 — Project Summary Report

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Project Overview

This project builds a machine learning regression model to predict house prices based on property features such as area, number of rooms, amenities, and location. The dataset contains 545 records and 13 features. Two models were trained and compared — Linear Regression and Random Forest Regressor — and the results were evaluated using MAE, RMSE, and R² Score.

Model Performance Results

Metric	Linear Regression	Random Forest
MAE	■2,16,262	■2,66,188
RMSE	■2,69,251	■3,43,080
R ² Score	0.9386 (93.9%)	0.9002 (90.0%)
Winner	✓ Better	—

Key Findings & Insights

- Most Influential Features:** The top factors driving house price are area (plot size), preferred area location, air conditioning, number of bedrooms, and number of bathrooms. Area alone accounts for the largest share of price variation.
- Model Accuracy:** Linear Regression achieved an R² of 0.9386, meaning it explains 93.9% of price variation. On average, predictions are within ■2.16 lakh of the actual price — a strong result for a dataset of this size and complexity.
- Surprising Finding:** Air conditioning had a stronger price impact than the number of bedrooms, highlighting that buyers place high value on comfort amenities. The preferred area premium was also larger than expected, confirming that location is a dominant factor.
- Business Recommendation:** Real estate businesses should prioritize marketing properties in preferred areas with air conditioning and large floor area. This model can also be used to identify underpriced listings — properties where the predicted price significantly exceeds the asking price represent strong investment or quick-sale opportunities.

Tools Used: Python 3 | Pandas | Scikit-learn | Matplotlib | Seaborn | Jupyter Notebook